

ECON 5345 Homework 4 Report

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Question 1

The FRED said that the TWEXBPA is no longer updated since some time around 2015, and it started to use RTWEXBGS “Real Broad Dollar Index” to substitute. So I used the RTWEXBGS instead. The following table reports the test statistics and critical values for the unit root tests on log levels.

Table 1: Unit root tests on log levels (test statistic and critical values).

Series	Test (spec.)	Stat	CV(1%)	CV(5%)	CV(10%)
<i>Panel A: log(dollar index), monthly (N=242)</i>					
RTWEXBGS	PP (Z- τ , const.)	-0.9725	-3.4589	-2.8736	-2.5731
RTWEXBGS	PP (Z- τ , trend)	-2.8287	-3.9993	-3.4297	-3.1381
RTWEXBGS	ADF (τ_2 , drift)	-1.2374	-3.4600	-2.8800	-2.5700
RTWEXBGS	ADF (τ_3 , trend)	-3.2276	-3.9900	-3.4300	-3.1300
RTWEXBGS	KPSS (μ)	3.7950	0.7390	0.4630	0.3470
RTWEXBGS	KPSS (τ)	0.5219	0.2160	0.1460	0.1190
<i>Panel B: log(real GDP), quarterly (N=316)</i>					
GDPC1	PP (Z- τ , const.)	-2.2491	-3.4528	-2.8709	-2.5717
GDPC1	PP (Z- τ , trend)	-1.3899	-3.9908	-3.4256	-3.1357
GDPC1	ADF (τ_2 , drift)	-2.3118	-3.4400	-2.8700	-2.5700
GDPC1	ADF (τ_3 , trend)	-1.5289	-3.9800	-3.4200	-3.1300
GDPC1	KPSS (μ)	5.3256	0.7390	0.4630	0.3470
GDPC1	KPSS (τ)	1.0925	0.2160	0.1460	0.1190

For the dollar index, the Phillips-Perron test fails to reject the null of unit root at 10% significance for both drift model and trend model. The augmented Dickey-Fuller test fails to reject the null of unit root at 10% significance for the drift model and rejects the null of unit root at 10% significance for the trend model. The KPSS test rejects the null of both level and trend stationarity at 1% significance.

For the real GDP, the Phillips-Perron test fails to reject the null of unit root at 10% significance for both drift model and trend model. The augmented Dickey-Fuller test rejects the null of unit root at 10% significance for the drift model, and fails to reject the null of unit root at 10% significance for the trend model. The KPSS test rejects the null of both level and trend stationarity at 1% significance.

Results from different tests can vary, and results can be different for the same test with different specifications. In this case, we may do various tests to get a better understanding of the data, and choose whether to reject unit root based on a majority rule.

Question 2

a. Note that we can write

$$\text{Var}(e) = [\Gamma_{ij}]_{i,j=1}^n \in \mathbb{R}^{Tn \times Tn},$$

where $\Gamma_{ij} = \text{Cov}(e_i, e_j) \in \mathbb{R}^{T \times T}$ is the (i, j) -th block of $\text{Var}(e)$. Since v_t is white noise, we have

$$\text{Cov}(e_{it}, e_{j\tau}) = \begin{cases} \Omega_{ij} \in \mathbb{R}, & t = \tau \\ 0, & \text{otherwise} \end{cases}$$

Therefore, we have $\Gamma_{ij} = \text{diag}(\Omega_{ij}) = \Omega_{ij} I_T$. Hence,

$$\text{Var}(e) = \Omega \otimes I_T.$$

b. Since Ω is a covariance matrix, it is positive definite. Hence, Ω is non-singular. By the properties of Kronecker product, we have

$$\begin{aligned} \hat{\beta}_{\text{GLS}} &= ((I_n \otimes X')(\Omega^{-1} \otimes I_T)(I_n \otimes X))^{-1} ((I_n \otimes X')(\Omega^{-1} \otimes I_T)y) \\ &= (\Omega^{-1} \otimes X'X)^{-1} (\Omega^{-1} \otimes X')y \\ &= (\Omega \otimes (X'X)^{-1}) (\Omega^{-1} \otimes X')y \quad \text{since } X'X \text{ is positive-definite} \\ &= (I_n \otimes (X'X)^{-1} X') y \\ &= (I_n \otimes (X'X)^{-1}) (I_n \otimes X')y \\ &= (I_n \otimes (X'X)^{-1}) ((I_n \otimes X)'y) \\ &= ((I_n \otimes X)'(I_n \otimes X))^{-1} ((I_n \otimes X)'y) \\ &= \hat{\beta}_{\text{OLS}}. \end{aligned}$$

Question 3

a. The values of BIC, AIC, and LRT up to $p = 10$ are reported in the following table.

Table 2: Lag selection for VAR (AIC/BIC) and sequential LR test p -values for $H_0 : p - 1$ vs $H_1 : p$.

p	AIC	BIC	LR p -value
0	1982	1992	N/A
1	1323	1363	7.06×10^{-140}
2	1301	1371	8.00×10^{-6}
3	1270	1370	1.38×10^{-7}
4	1278	1409	4.08×10^{-1}
5	1266	1426	3.25×10^{-4}
6	1261	1451	6.88×10^{-3}
7	1264	1484	7.86×10^{-2}
8	1277	1528	8.97×10^{-1}
9	1283	1564	2.23×10^{-1}
10	1265	1575	3.29×10^{-5}

The AIC selects $p = 6$, the BIC selects $p = 1$, and the LRT selects $p = 3$.

b. Least squares estimation results of the VAR(6) model are reported in Table 3.

Table 3: VAR(6) least squares estimates. Each cell reports the coefficient (with significance stars) and the t -statistic in parentheses.

RHS regressor	$\Delta \log(GDP_t)$	$\Delta \log(CPI_t)$	FFR_t
	0.2213**	0.0717	0.3751***
$\Delta \log(GDP_{t-1})$	(2.890)	(1.765)	(5.171)
	-0.0783	0.4416***	-0.0140
$\Delta \log(CPI_{t-1})$	(-0.465)	(4.947)	(-0.088)
	0.0638	0.1342**	1.1156***
FFR_{t-1}	(0.809)	(3.208)	(14.937)
	0.1430	-0.0254	0.1976**
$\Delta \log(GDP_{t-2})$	(1.809)	(-0.606)	(2.639)
	-0.1540	0.0307	0.4786**
$\Delta \log(CPI_{t-2})$	(-0.828)	(0.311)	(2.717)
	-0.3680**	-0.1069	-0.5345***
FFR_{t-2}	(-3.243)	(-1.775)	(-4.972)
	0.0018	0.0531	0.0990
$\Delta \log(GDP_{t-3})$	(0.022)	(1.236)	(1.292)
	-0.0508	0.4392***	0.0231
$\Delta \log(CPI_{t-3})$	(-0.271)	(4.409)	(0.130)
	0.3456**	0.0114	0.4932***
FFR_{t-3}	(2.862)	(0.178)	(4.312)
	0.0684	0.0625	-0.0408
$\Delta \log(GDP_{t-4})$	(0.886)	(1.524)	(-0.557)
	-0.0026	-0.1674	-0.2671
$\Delta \log(CPI_{t-4})$	(-0.014)	(-1.671)	(-1.494)
	-0.1462	-0.0637	-0.1994
FFR_{t-4}	(-1.194)	(-0.980)	(-1.718)
	-0.0541	-0.0076	0.1830*
$\Delta \log(GDP_{t-5})$	(-0.690)	(-0.182)	(2.463)
	0.3090	0.1017	0.1916
$\Delta \log(CPI_{t-5})$	(1.667)	(1.034)	(1.091)
	-0.0211	0.1034	0.2911*
FFR_{t-5}	(-0.176)	(1.619)	(2.553)
	0.0778	0.0145	-0.0369
$\Delta \log(GDP_{t-6})$	(1.030)	(0.362)	(-0.516)
	-0.0898	0.0403	0.1392
$\Delta \log(CPI_{t-6})$	(-0.523)	(0.442)	(0.856)
	0.1043	-0.0738	-0.2776***
FFR_{t-6}	(1.283)	(-1.712)	(-3.605)
	0.6122**	-0.0568	-0.5384*
Constant	(2.789)	(-0.487)	(-2.589)

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1. t -values in parentheses.

- c. The impulse responses are reported in the Table 4 with functions plotted with error bands in Figure 1. The Monte Carlo results and bootstrap results are plotted together, with blue bands for MonteCarlo and pink bands for bootstrap. Overlapping bands are colored in grey.

Table 4: Impulse responses to a one-standard-deviation shock in FFR (Cholesky ordering $[\Delta \log GDP, \Delta \log CPI, FFR]$). Responses of $\log(GDP)$ and $\log(CPI)$ are obtained by cumulating the responses of $\Delta \log(GDP)$ and $\Delta \log(CPI)$.

Horizon h	$\log(GDP)$	$\log(CPI)$	FFR
0	0.0000	0.0000	0.7243
1	0.0462	0.0972	0.8081
2	-0.1663	0.1745	0.5303
3	-0.2410	0.1882	0.4918
4	-0.2962	0.2141	0.4929
5	-0.4245	0.2762	0.5219
6	-0.4549	0.3013	0.4504
7	-0.4597	0.3049	0.2734
8	-0.4868	0.2972	0.2152
9	-0.4665	0.2717	0.1954
10	-0.4433	0.2604	0.1503
11	-0.4196	0.2457	0.1108
12	-0.3829	0.2128	0.0686
13	-0.3552	0.1898	0.0534
14	-0.3321	0.1703	0.0498
15	-0.3073	0.1470	0.0352
16	-0.2871	0.1308	0.0293
17	-0.2719	0.1143	0.0229
18	-0.2602	0.0972	0.0169
19	-0.2498	0.0854	0.0166
20	-0.2421	0.0734	0.0107

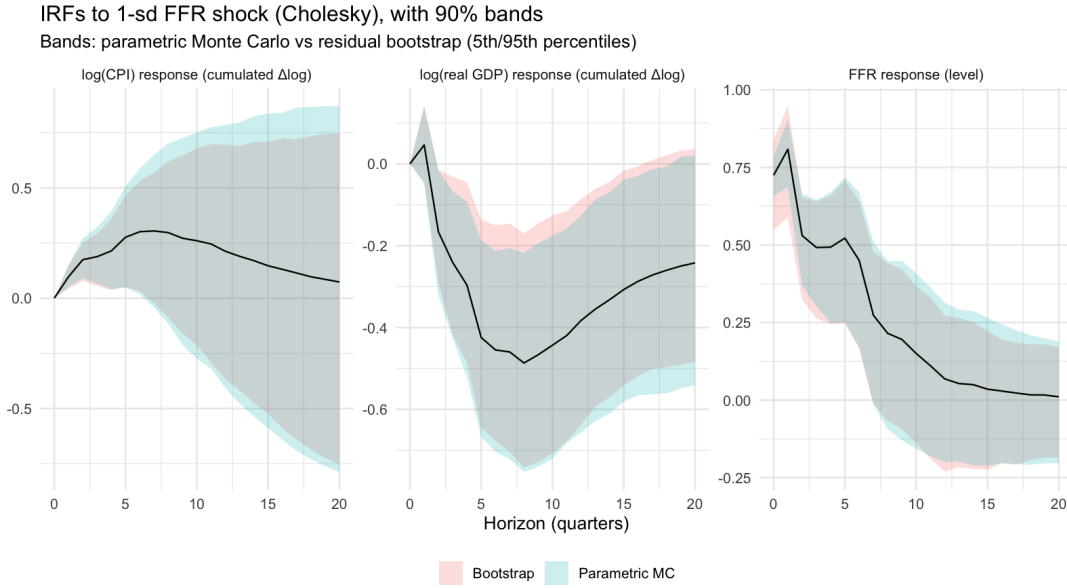


Figure 1: Impulse Responses to FFR Shock with Error Bands

The price level first increases and then decreases, revealing a hump-shape. This is referred to as the "price puzzle". The Fed reacts to the inflation expectation for a change in the FFR. The VAR here only includes three variables, which is not enough to capture the full set of variables or information used by the Fed.

To solve the problem, we need to include commodity prices and other asset prices in the VAR. They are forward-looking variables, which, to some degree, capture Fed's information about future macro fundamentals.

Barth & Ramey (2002) introduces the "cost channel of monetary policy", arguing that when the FFR increases, the cost of holding inventories increases, which requires a rise in price level to compensate for firms' marginal cost.

The FEVD is plotted in Figure 2.

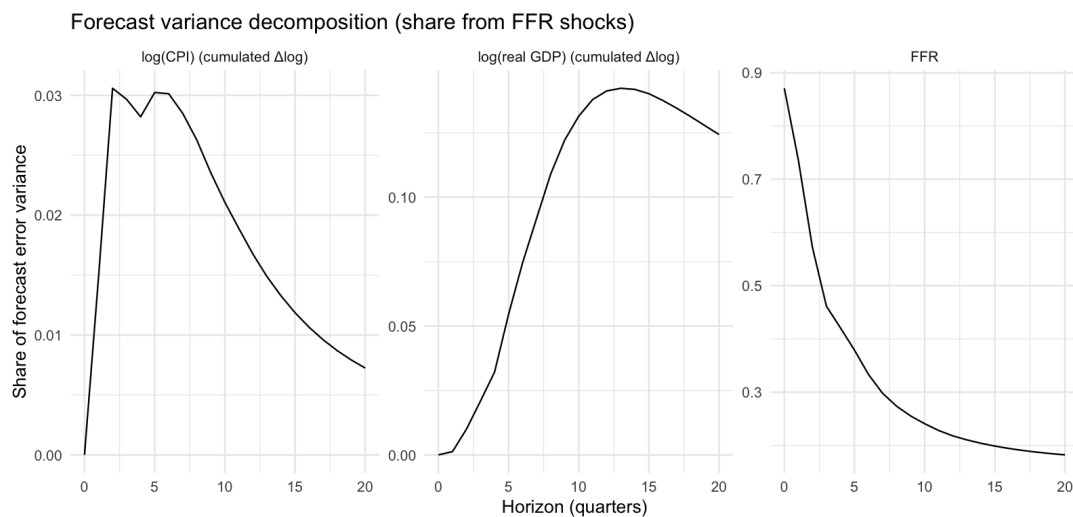


Figure 2: FEVD of FFR Shock